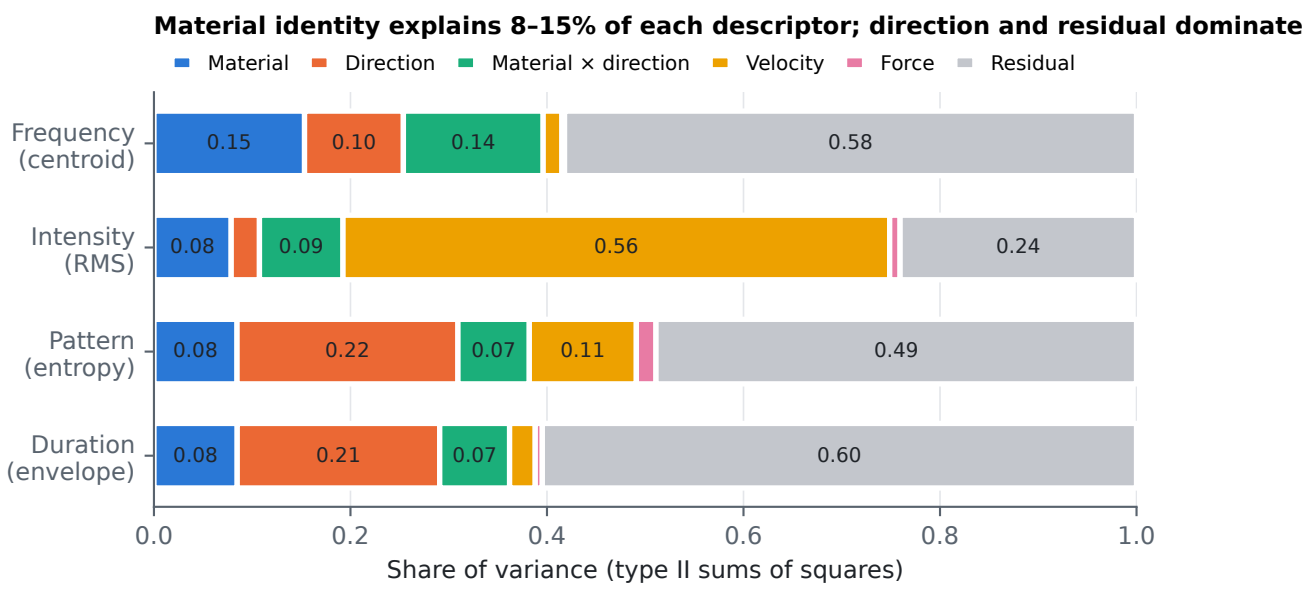
Material identity explains 8–15% of each descriptor; direction and residual dominate
Material
Direction
Material × direction
Velocity
Force
Residual
Frequency (centroid)
Intensity (RMS)
Pattern (entropy)
Duration (envelope)
0.15
0.10
0.14
0.58
0.08
0.09
0.56
0.24
0.08
0.22
0.07
0.11
0.49
0.08
0.21
0.07
0.60
0.0
0.2
0.4
0.6
0.8
1.0
Share of variance (type II sums of squares)